

SIDDHI LAD

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EDUCATION

The Pennsylvania State University | University Park, PA August 2024 – May 2026
Master of Science in Cybersecurity Analytics and Operations GPA – 3.97

NIIT University | Rajasthan, India August 2018 – July 2022
Bachelor of Technology in Computer Science and Engineering GPA – 3.21

WORK EXPERIENCE

BSM | *Cybersecurity Engineer Intern* | Mumbai, India June 2025 – August 2025

- Built a security automation tool in Python using Regex, Pandas and Streamlit to parse WatchGuard firewall XML configuration and detect policy violations, cutting drift auditing time by 50%.
- Deployed a remote device management framework using Ansible and PowerShell to capture secure system baselines, perform automated deviation detection, and generate real-time host telemetry alerts.
- Performed post-incident digital forensic analysis on a Windows workstation compromised by a ransomware; parsed Windows Event Logs and Registry hives to map malicious privilege escalation and isolate anti-forensic activities including Volume Shadow Copy subversion.
- Conducted security gap analysis against existing incident response procedures and international regulatory standards (IMO 2021/ISO 27001); pitched data-driven remediation workflows to the company CISO including automated metric collection and vendor SLA tracking.

Penn State University | *Instructional Assistant* | University Park, PA August 2024 – May 2026

- Supported cybersecurity courses (Network Security and Malware Analytics) for 50+ students; guided technical labs covering network traffic analysis, TCP/IP protocols, SSL security, and IDS/IPS (Zeek) configuration.
- Instructed students on reverse engineering and static/dynamic malware analysis techniques using professional tools including Ghidra, IDA Pro, x64dbg, and GDB to identify indicators of compromise (IOCs).

IBM | *Information Security Analyst* | Bangalore, India January 2022 – May 2024

- Automated stateful pre/post-deployment configuration diffing of Cisco ASA firewalls using Python, Secure SSH (Paramiko), and regex-based log parsing, cutting down manual efforts by 60% and ensuring reliable deployments.
- Secured 1000+ HIPAA-compliant web applications using Imperva Cloud WAF for a healthcare client; configured and tuned custom rules to mitigate OWASP Top 10 vulnerabilities including SQL injection, XSS and DDoS.
- Investigated and remediated 200+ incidents ranging from P1-P4 severity levels via traffic and log analysis, rate limiting, and IP/Geo-blocking; served as the technical POC for leadership to communicate RCAs, remediation timelines, and strategic risk updates to cross-functional stakeholders.
- Managed SSL certificate renewals and TLS cipher suite updates by collaborating with vendor, application, and DNS teams; configured load balancing and enforced role-based access control policies (RBAC).
- Optimized enterprise Change Management performance by auditing ITSM ticket routing within ServiceNow, implementing refined Standard Operating Procedures (SOPs) that reduced workflow latency from 15 days to 3 days.

TECHNICAL SKILLS

Scripting – C/C++, Python, PowerShell, SQL | **Web-Development** – HTML, CSS, JavaScript, Django, Flask | **Security Engineering** – Threat Modeling, Secure Code Review, Security Automation, API Security, SDLC | **DevOps** - CI/CD, Ansible, Docker, Jenkins, Git/GitHub, AWS (VPC, EC2, S3) | **Network Security** – Firewall Management, Imperva WAF, Cisco ASA, DNS, TCP/IP, LDAP, SMTP | **SIEM** – Splunk, ELK, Wazuh | **OS** – Linux, Windows | **Certifications** – AWS Certified Cloud Practitioner (CCP), CompTIA Security+, Certified Ethical Hacker (CEH v11)

PROJECTS

Cascading Risks in Open-Banking APIs | *Threat Modeling, OAuth/OIDC, API Security* January 2026

- Designed and deployed a containerized microservices testbed to model systemic risk propagation in Open Banking architectures (FastAPI, Kong, Keycloak, ELK, PostgreSQL, Docker); applied the STRIDE threat modeling framework and simulated API exploits to validate existing security controls.

IDS for IoT Networks using ML | *LSTM, GNN, MITRE ATT&CK* September 2025

- Developed an IDS to classify IoT network attacks using machine learning (LSTM, GNN, and Logistic Regression) on the UNSW-NB15 dataset and mapped intrusion attempts to MITRE ATT&CK tactics, techniques and procedures (TTPs) for structured reporting.
- Explored multi-model consensus which acts as a high-precision filter, boosting accuracy of attack classification from 1.82% to 27.05% when models agree, useful for producing explainable threat detection.

COMPETITIONS AND COMMUNITY INVOLVEMENT

The Diana Initiative (TDI) | *Volunteer* May 2026

- Supported conference operations by engaging with attendees, facilitating speaker sessions, and connecting participants with relevant resources across cybersecurity talks.

Wicked6 Cyber Games | *4th Place out of 10 teams* March 2026

- Analyzed source code to identify and exploit vulnerabilities including PHP type juggling to bypass authentication, SCP path traversal in Go based SSH service, and buffer overflow attacks over a compiled C binary.

Hack The Box – Red & Blue CTF | *Top 10 Finish out of 20+ teams* March 2026

- Solved web and network forensic challenges by exploiting command and SQL injection vulnerabilities, analyzing DNS tunneling and data exfiltration via encoded subdomains as well as reconstructing SMTP-based Base64 MIME email attachments from PCAP traffic using Wireshark.

Blue Team, Pros vs Joes, BSides NYC | *1st Place out of 4 teams* October 2025

- Defended Windows endpoints against live red team attacks by hardening configurations using Active Directory and Group Policy enforcement to restrict and prevent lateral movement.

NCAE-C Cyber Games | *Top 5 Finish out of 12 teams* February 2025

- Configured nginx server and maintained the availability of web service during active attack by patching found vulnerabilities; solved web exploitation, network forensics and cryptography CTF challenges.